№1.

$$G = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$$+\begin{pmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix} = [I_{3\times 3} \ P]$$

$$H = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{pmatrix} \Leftarrow H_{sys} = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

Хар-ки: $R = \frac{k}{n} = \frac{3}{5}$, $d_{min} = 1$, $t = 0$

код. слова:

```
[0, 0, 0, 0, 0]
[1, 0, 1, 1, 0]
[0, 1, 1, 0, 1]
[1, 1, 0, 1, 1]
[1, 1, 0, 1, 0]
[0, 1, 1, 0, 0]
[1, 0, 1, 1, 1]
[0, 0, 0, 0, 1]
```

таблица синдромного декод:

s	e
00	00000
01	00100
10	10000
11	00010

```
G = [[1, 1, 0, 1, 0],
     [0, 1, 1, 0, 1],
     [1, 0, 1, 1, 0]]

H = [[1, 0, 0, 1, 0],
     [0, 1, 1, 1, 0]]

codeWords = []
d_min = 10000
for i in range(2):
  for j in range(2):
    for k in range(2):
      tmp = []
      for p in range(5):
        tmp.append((G[0][p] * i + G[1][p] * j + G[2][p] * k) % 2)
      codeWords.append(tmp)

for i in range(len(codeWords)):
  for j in range(len(codeWords)):
    if i == j: continue
    z = 0
    for u in range(5):
      if codeWords[i][u] != codeWords[j][u]:
        z += 1
    d_min = min(d_min, z)

for i in codeWords:
  print(i)

print()
print(f'd_min={d_min}')
```

```
from itertools import product

def prdct(a, B):
  res = []
  for i in range(len(B)):
    tmp = 0
    for j in range(len(a)):
```

```
      tmp = (tmp + a[j] * H[i][j]) % 2
    res.append(tmp)
  return res

es = list(product([0, 1], repeat=5))
s = [prdct(e, H) for e in es]
decodeWords = [[1, 1, 1, 1, 1], [1, 1, 1, 1, 1], [1, 1, 1, 1, 1], [1, 1,
1, 1, 1]] # 00, 01, 10, 11
for i in range(len(es)):
  if (s[i][0] == 0) and (s[i][1] == 0):
    if sum(es[i]) < sum(decodeWords[0]):
      decodeWords[0] = es[i]

  if (s[i][0] == 0) and (s[i][1] == 1):
    if sum(es[i]) < sum(decodeWords[1]):
      decodeWords[1] = es[i]

  if (s[i][0] == 1) and (s[i][1] == 0):
    if sum(es[i]) < sum(decodeWords[2]):
      decodeWords[2] = es[i]

  if (s[i][0] == 1) and (s[i][1] == 1):
    if sum(es[i]) < sum(decodeWords[3]):
      decodeWords[3] = es[i]

print(decodeWords)
```

№2. $(\overset{n}{23},\overset{k}{12})$

A parity check matrix for the binary Golay code is given by the matrix $H = (M\ I_{11})$, where I_{11} is the 11×11 identity matrix and M is the 11×12 matrix

$$M = \begin{bmatrix} 1&0&0&1&1&1&0&0&0&1&1&1 \\ 1&0&1&0&1&1&0&1&1&0&0&1 \\ 1&0&1&1&0&1&1&0&1&0&1&0 \\ 1&0&1&1&1&0&1&1&0&1&0&0 \\ 1&1&0&0&1&1&1&0&1&1&0&0 \\ 1&1&0&1&0&1&1&1&0&0&0&1 \\ 1&1&0&1&1&0&0&1&1&0&1&0 \\ 1&1&1&0&0&1&0&1&0&1&1&0 \\ 1&1&1&0&1&0&1&0&0&0&1&1 \\ 1&1&1&1&0&1&0&0&1&1&0&1 \\ 1&1&1&1&0&0&0&0&1&1&0&1 \\ 0&1&1&1&1&1&1&1&1&1&1&1 \end{bmatrix} \begin{matrix} 1 \\ \ddots \\ \vdots \\ 0 \\ 0 \\ 1 \end{matrix}$$

мин. зав. столбцов (7 штук)

$\forall$ 6 столбцов мин. нез., а 7-мин зав.

$\hookrightarrow d_{min} = 7 \Rightarrow t = 3$

$$M \le \frac{q^n}{\sum_{i=0}^{t} C_n^i (q-1)^i}; \quad M = 2^k = 2^{12}$$

$$\frac{2^{23}}{1 + \frac{23!}{22!} + \frac{23!}{2 \cdot 21!} + \frac{23!}{6 \cdot 20!}} =$$

$$= \frac{2^{23}}{\underbrace{1 + 23 + 253 + 1771}_{\text{объем сфер}}} = \frac{2^{23}}{2^{11}} = 2^{12} = M \Rightarrow$$ все слова в сферах, чтд

```python
from itertools import combinations


H = [[1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0],
     [1, 0, 1, 0, 1, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0,
0],
     [1, 0, 1, 1, 0, 1, 1, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,
0],
     [1, 0, 1, 1, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,
0],
     [1, 1, 0, 0, 1, 1, 1, 0, 1, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,
0],
     [1, 1, 0, 1, 0, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0,
0],
     [1, 1, 0, 1, 1, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0,
0],
     [1, 1, 1, 0, 0, 1, 0, 1, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0,
0],
     [1, 1, 1, 0, 1, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0,
0],
     [1, 1, 1, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1,
0],
     [0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
1]]

count = 0
dd = 6 # число столбцов для суммирования
coefs = [[1 if i in pos else 0 for i in range(23)] for pos in
combinations(range(23), dd)]
for coef in coefs:
  count += 1
  s = [0] * 23
  for i in range(23):
    if coef[i] == 0:
      continue
    for j in range(11):
      s[j] = (s[j] + H[j][i]) % 2

  if sum(s) == 0:
    print(coef)
    print(s)
    break
```